

T.SVD - Spring Validation Demonstration Test	
Objective	Perform an experiment semi-autonomously
Elements	User Interface subsystem, perception subsystem, planning subsystem, manipulation subsystem, lab machine integration subsystem, monitoring subsystem.
Location	Mellon Institute
Equipment	Computer, XArm 6, cameras, OpenTron OT-2, custom shaker module, well plates.
Personnel	Arnav, Farnaz, Karthik, Juan
Procedure	☐ A user enters an experiment description. ☐ The XArm 6 scans the environment to detect necessary equipment. ☐ The perception subsystem detects, segments, and localizes the desired equipment. It passes this information to the planning subsystem. ☐ The planning subsystem generates a plan to grasp, move, and place the desired equipment in the OT-2. The manipulation subsystem executes this plan. ☐ A team member presses the equipment into the OT-2 slot. ☐ The OT-2 is commanded to dispense liquid from a test tube rack that has already been filled and placed inside of the OT-2. ☐ The planning subsystem generates a plan to grasp, move, and place the desired equipment in the shaker module. The manipulation subsystem executes this plan. ☐ The shaker module is commanded to shake for 30 seconds. ☐ The planning subsystem generates a plan to grasp, move, and place the desired equipment in the output bin. The manipulation subsystem executes this plan.
Verification Criteria	☐ The system completes the experiment end-to-end. ☐ Grasp and movement of the desired equipment is performed within a 5mm error bound, 75% accuracy ☐ Path planning >95% of the time ☐ Detection accuracy >90%